Joohan Bae

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RESEARCH INTERESTS

Human–AI interaction; interaction techniques that help users and generative AI align intent and interpretation; visualizing how AI interprets instructions for non-expert users; human–robot interaction

EDUCATION

Kogakuin University, Tokyo, Japan

M.Eng. in Mechanical Engineering — Apr 2026 – Present

- Advisor: Prof. Daigo Misaki (Misaki Laboratory)
- Early completion under faculty review; expected Mar 2027 if approved
- GPA: 3.79/4.0 (16 credits completed as of Oct 2026)

B.Eng. in Mechanical Systems Engineering — Apr 2020 – Mar 2026

- GPA: 3.6/4.0
- Leave of absence for mandatory military service, Republic of Korea (Apr 2022 – Mar 2024)

PUBLICATIONS

International conference papers

1. Daigo Misaki, **Joohan Bae**, and Komei Miura. 2026. Designing Human–AI Interaction to Foster Ambidextrous Thinking and Learner Ownership in Product Ideation. In *AHFE 2026 Hawaii International Conference: Human Factors in Design, Engineering, and Computing*, Honolulu, HI, USA, Dec 2026. To appear.

Peer-reviewed posters

2. **Joohan Bae** and Daigo Misaki. 2026. Who Pays for Vagueness? A Visual Token-Metered Interaction Technique for Promoting Precise AI Commands with TokenTangram. In *Adjunct Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology (UIST '26 Adjunct)*. Poster. To appear. doi:10.1145/3830397.3841836
3. Yuya Aikawa, **Joohan Bae**, and Daigo Misaki. 2025. Clay Play: A Metaphorical LLM Interface for Dynamic Idea Generation and Exploration. In *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25 Adjunct)*, Busan, Republic of Korea. Poster. doi:10.1145/3746058.3758345

Under review

4. Full paper on how users form and revise capability beliefs about an LLM-enabled tabletop robot (second author). *ACM/IEEE International Conference on Human-Robot Interaction (HRI '27)*. Under review; details withheld during double-blind review.
5. **Joohan Bae**, Victor Taratukhin, and Daigo Misaki. Converging Through Speech or Through AI-Refined Text: Four Team Cases from a Hands-On Design Thinking Seminar for Engineering Students. *Journal of Robotics and Mechatronics*. Under revision.

PRESENTATIONS

- Daigo Misaki and **Joohan Bae**. 2026. Integrating Vibe Coding into Mechanical Systems Engineering Education: From Mechatronics Experiments to Prototyping in Undergraduate Thesis Research. 25th International Symposium on Advanced Technology (ISAT-25), Southern Taiwan University of Science and Technology, Tainan, Taiwan, Oct 2026. Poster.
- Nana Sakai, Mayu Ikeda, **Joohan Bae**, Genki Sakai, Daigo Misaki, and Victor Taratukhin (equal contribution). 2026. Prototype of a Conversational Robot with Facial Expressions and Body Movements for Sustained Dialogue with Older Adults (in Japanese). 12th Medicine–Pharmacy–Engineering Three-University Symposium, Tokyo, Japan, Oct 2026. Poster.

RESEARCH EXPERIENCE

Master's Research, Misaki Laboratory, Kogakuin University Apr 2026 – Present

- Thesis: *Interaction Support for Meaning Negotiation in Human–Generative AI Collaboration*. Built TokenTangram, which visualizes how users and AI negotiate intent by metering tokens per instruction and replaying how the AI interpreted each one (UIST '26 poster, first author)
- Planning a 12-participant within-subjects study (Dec 2026) testing whether billing instruction cost by length or by attempt changes users' awareness and effort allocation

Kogakuin University × University of Münster Joint Project-Based Learning 2026

- Led circuits, firmware, and interaction design of IkigAI, a communication robot for older adults, in a 15-member Japanese–German team, and coordinated the two teams in English
- Ran a 10-participant human–robot interaction study with the robot (HRI '27, under review)

Design Thinking Workshop Study, Misaki Laboratory

- Compared how four teams (16 engineering students) converged on ideas through speech or AI-refined text; led the analysis and first-authored the journal manuscript

TEACHING EXPERIENCE

Teaching Assistant, Design Engineering, Kogakuin University Sep 2026 – Present

SCHOLARSHIPS

- Kogakuin University Scholarship for Privately Financed International Students with Outstanding Academic Performance — 2025, 2026
- Taisei Oncho Scholarship Foundation Scholarship (two-year award) — 2025 – 2026
- Monbukagakusho Honors Scholarship for Privately-Financed International Students, Japan Student Services Organization (JASSO) — 2021, 2024

LEADERSHIP AND ACTIVITIES

Kogakuin Robot Project (KRP) 2020 – 2022

- Developed robot control software; led the programming team (2021 – 2022)

SKILLS

- **Programming:** Python, C, JavaScript
- **Hardware prototyping:** Raspberry Pi, Arduino, M5Stack; sensor and actuator control with basic circuits
- **CAD:** Autodesk Fusion
- **Research methods:** User studies, questionnaires, and interviews; statistical analysis of questionnaire data (ANOVA, MANOVA); MATLAB

LANGUAGES

Korean (native) · Japanese (fluent) · English (professional working proficiency)

Last updated October 2026